POLITECNICO
DI MILANO

Laboratory of Nearable Technologies for Health Data Science

Real-Time Motion-Resilient rPPG

Supervisors:
Andrea Moglia
Marco Brando Mario Paracchini

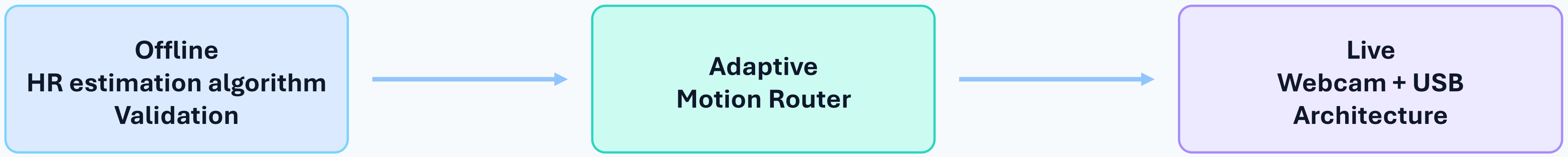
Team Members:

Asal Abbas Nejad Fard	10974178
Mohanesh Ravi	11035558
Mohammadreza Zamani	10869960

June 2026

Real-Time Motion-Resilient rPPG

Adaptive signal conditioning router with multi-threaded hardware validation



The Fundamentals of Remote Plethysmography

Problem : Signal-to-noise loss

Physics of rPPG

- Blood Volume Pulses (BVP)
- Backscattered light
- Remote Photoplethysmography
- The useful signal is weak compared with lighting and motion artifacts

Motion bottleneck

- Yaw, pitch, and rigid head movements alter specular and diffuse reflections
- When there are macro-level luminance changes it can dominate the biological AC/DC variation
- This results in poor SNR and unstable peak detection.

Objective

- Build a pipeline
- First Detect motion, then adapt signal conditioning
- Avoid over-filtering
- Build a pipeline that remains usable in real time

PROJECT WORKFLOW

System Innovation: Adaptive Signal Conditioning Router

Innovation : Routing before filtering

Core Algorithmic Architecture & System Pipeline Roadmap

Raw Signal Ingestion
RGB Face streams

→

Pre-Routing Analysis
Initial motion classification

→

Context-Aware Signal Tuning
Adaptive processing path

→

Physiological Extraction
HR / RR outputs

Design logic

- Classify- first approach
- Motion is identified, then ROI limits, normalization, and filters adapt
- The router preserves clean signals

Two-stage roadmap

- Phase I: Offline mathematical verification
- Phase II: Real-time causal deployment

PHASE - I

Offline Algorithm Validation with Motion Routing

PHASE I

Controlled Offline Evaluation: Experimental Acquisition Protocol

Offline dataset → 9 × 3 × 2

9 unique subjects	**3** motion tasks	**2** repetitions per task	**54** standardized sessions	**30 s** duration per recording

Acquisition engine

- Custom synchronization scrips
- Each recording uses a shared acquisition baseline for fair comparison.
- The goal is reproducibility across subjects, runs, and motion profiles.

Why this design matters

- Repeated motion tasks separates random errors from systematic motion sensitivity.
- Static pose establishes the upper-bound of performance condition.
- Motion trials stress the adaptive conditioning logic .

The 3 Behavioral Motion Paradigms

Motion profiles → static, yaw, sinusoidal

1. Stable baseline

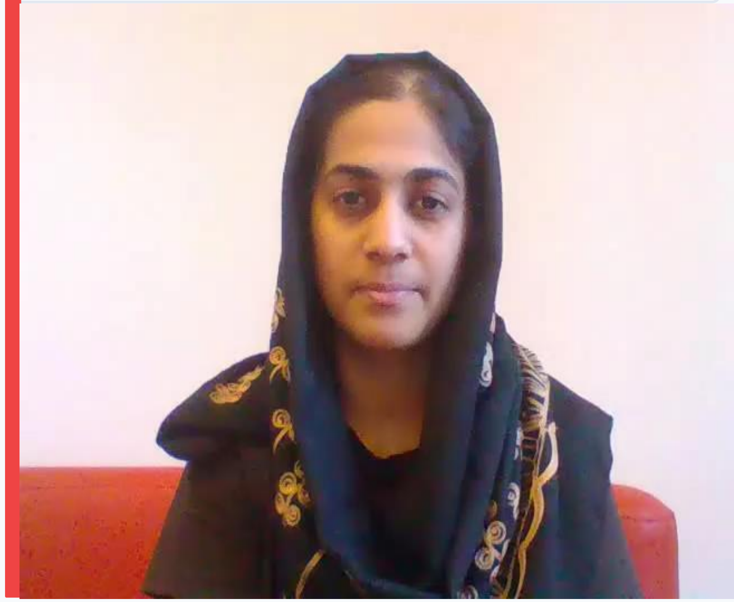
- Subject faces the camera
- Minimal head motion
- Reference condition (highest SNR)

2. Left-to-right rotation

- Continuous yaw motion
- Horizontal ROI displacement
- Moderate motion artifacts

3. Periodic sinusoidal motion

- Combined yaw + pitch motion
- Strong illumination variation
- Most challenging scenario



All recordings:
30 s duration | 30 FPS RGB video | Synchronized CMS50D acquisition

Motion Classification Results

The head-tracking module extracted horizontal and vertical motion trajectories and correctly classified the motion with a success rate of 100% for our custom created dataset

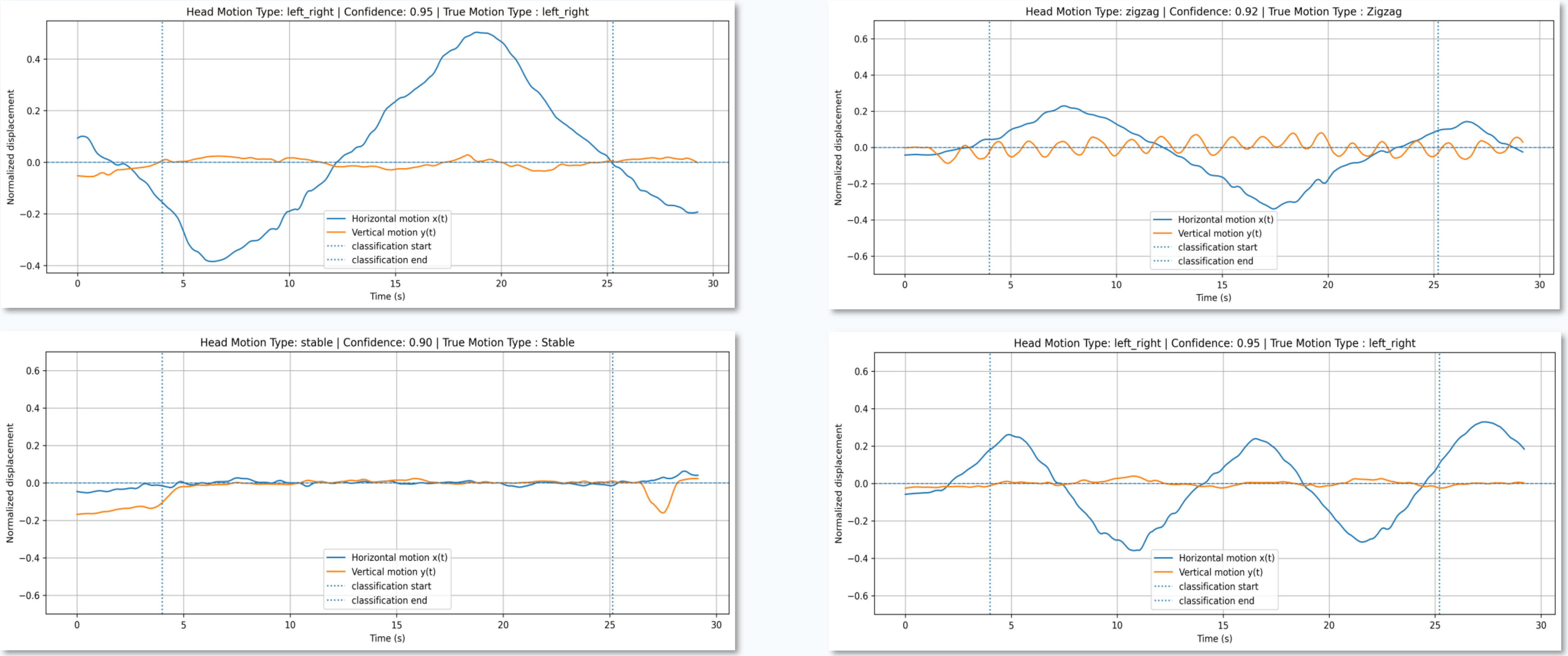


Figure 1: subset of results of motion detection algorithm results

OFFLINE PIPELINE

Algorithmic Blueprint for Offline Extraction

Video → RGB → BVP

Video frames → Face tracking + ROI mask → RGB Spatial pooling → Detrending / Filtering → Green /POS BVP signal

ROI Strategy

- Dynamic face tracking
- Multi-ROI skin extraction
- Forehead + bilateral cheeks

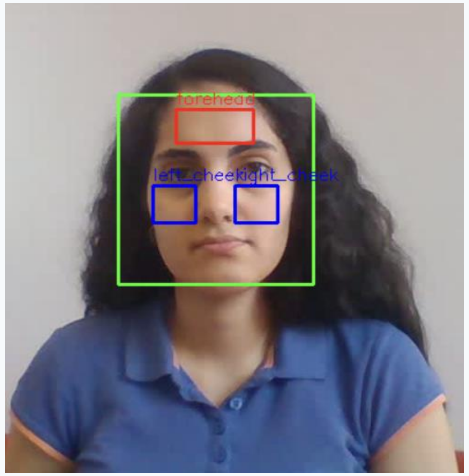


Figure 2: visual feedback of the ROI implemented

Signal Conditioning

- RGB spatial averaging
- Detrending
- Band-pass filtering
- Green /POS transformation

Blind Source Separation: POS and Green Models

RGB projection → physiological component

Both methods estimate the Blood Volume Pulse (BVP) from RGB signals .

POS model

- Orthogonal RGB projection
- Normalized skin-tone subspace
- More robust under motion

Green Channel

- Strongest hemoglobin absorption
- Highest pulsatile SNR
- Baseline BVP estimator
- sGreen(t)=G(t)

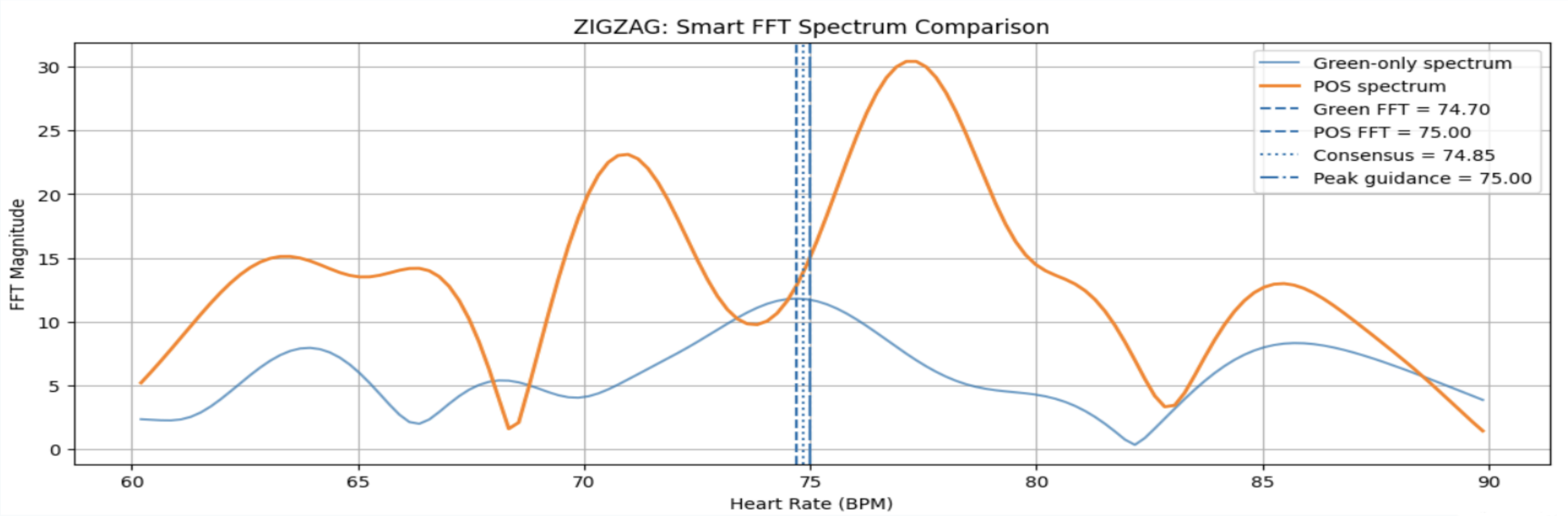


Figure 3: FFT Spectrum Comparison of POS and Green signal for a zigzag motion class video

Table 1: Comparison of Metrics

Metric	BPM
Green FFT	74.70
POS FFT	75.00
Consensus HR	74.85
CMS50D GT	77.00
MAE	2.15
MAPE	2.79 %

Spectral Extraction: Heart Rate Estimation

BVP → PSD peak → BPM

Filtered BVP time series → **Hanning window** → **Welch PSD over segments** → **Dominant peak in 1.0- 1.6 Hz (60- 100 BPM)**

HR window

1.0- 1.6 Hz
(60 - 100 BPM)

Heart Rate Conversion

HR = f_peak × 60

Peak Selection Rule

Highest PSD peak
within HR band

Experimental Example(Zigzag):

Motion : Zigzag
F_peak = 1.247 Hz
Estimated HR = 74.85 BPM
CMS50D GT = 77 BPM
MAE = 2.15 BPM
MAPE= 2.8%

Inference:

Selection of dominant spectral peak accurately recovered the reference heart rate.

Physiological Interfacing: Tachogram Construction

Systolic peaks → IBI → tachogram

Respiratory sinus arrhythmia

- We are exploiting the RSA in estimation of RR
- HR increases during inhalation
- HR decreases during exhalation
- Breathing modulates beat-to-beat intervals (IBI) in the Heart Rate

Peak finding

- Detect systolic peaks in filtered BVP
- Compute inter-beat intervals (IBI)

$$IBI_n = t_n - t_{n-1}$$

Tachogram signal

- Plot IBI values over time
- Tachogram acts as an indirect respiratory signal
- Used as input for RR estimation

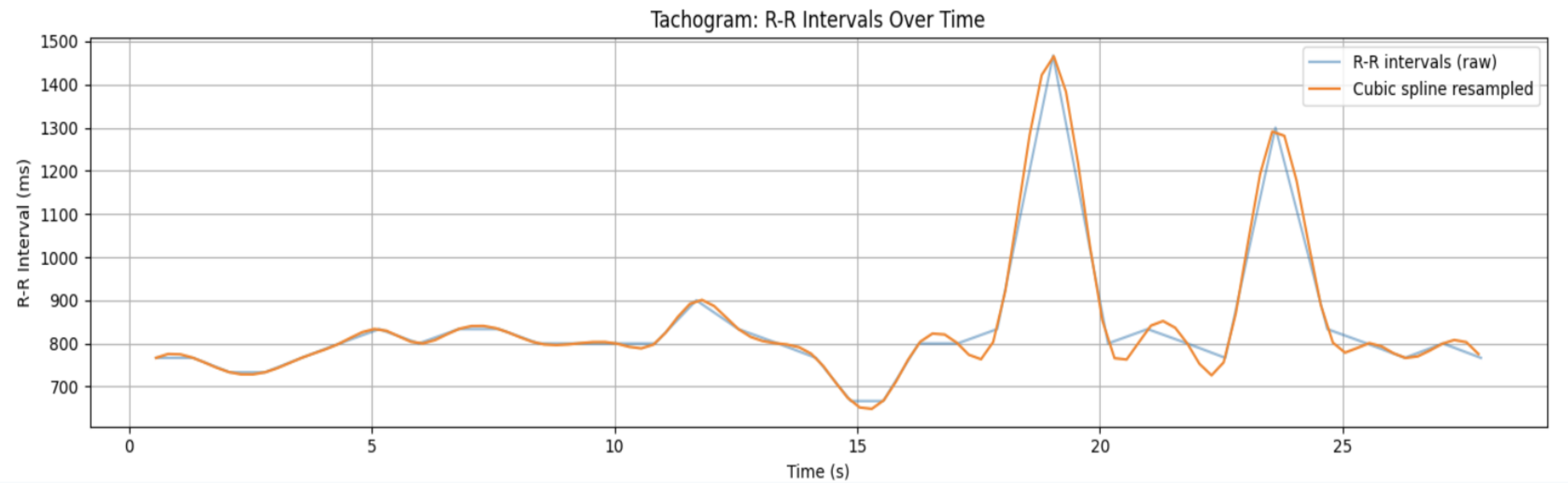


Figure 4 : Tachogram generated from the Zigzag recording.

- Estimated HR : 72.7 BPM
- Detected Peaks : 35 R-R
- Intervals : 34
- Mean IBI : 825.5 ms

Respiration Rate Estimation from the Tachogram

Irregular data → spline → low-frequency PSD

Irregular IBI samples → Cubic spline interpolation → Uniformly sampled breathing curve → PSD peak 9 - 36 Breaths/min

Why resampling?

- IBIs are irregularly spaced in time.
- Cubic spline reconstructs a continuous tachogram.

Respiration frequency band

- Search band: 0.15 - 0.6 Hz
- Equivalent: 9 - 36 breaths/min
- RR estimated from Dominant spectral peak

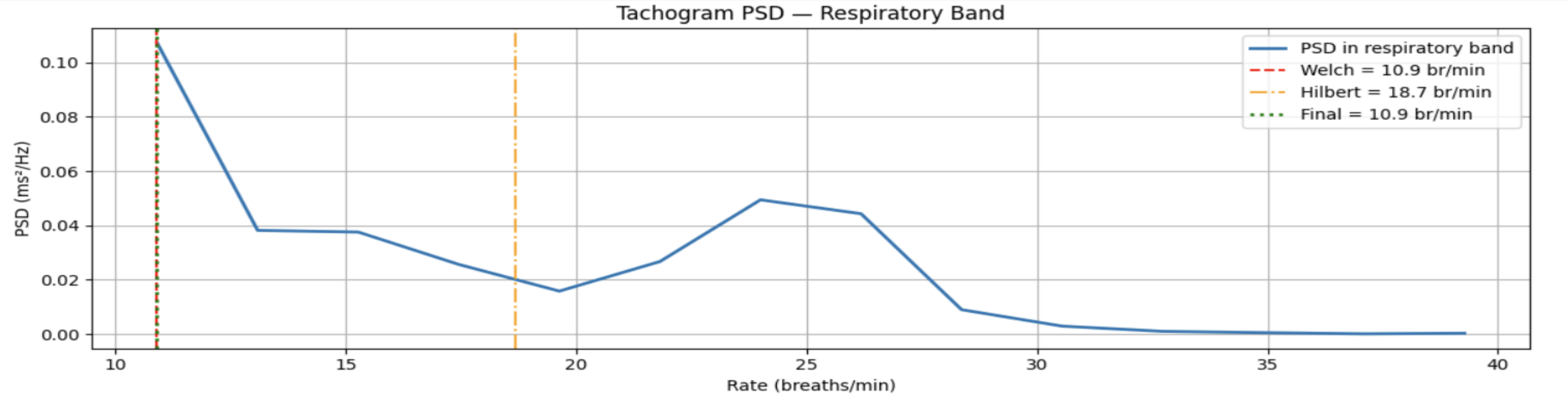


Figure 5 : Respiratory Rate estimation from the Tachogram PSD

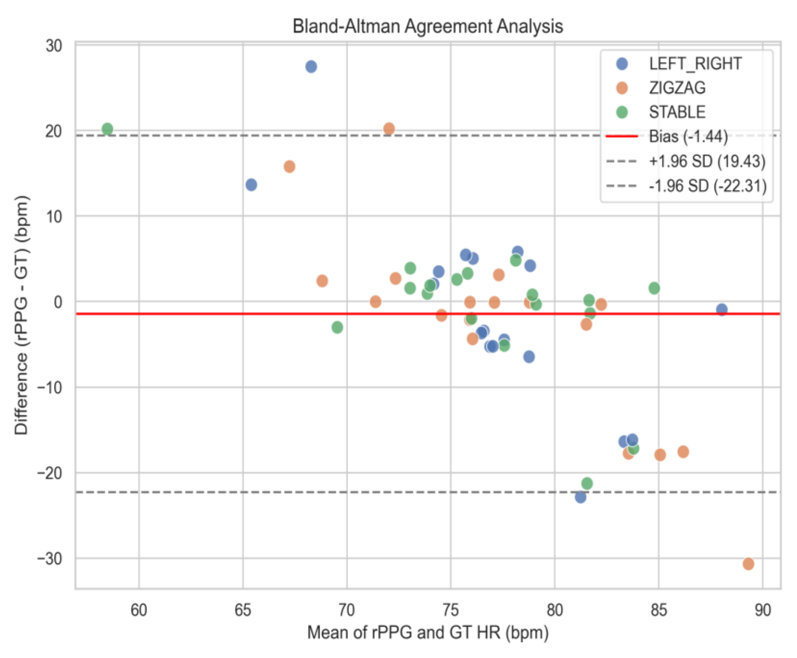
Overall Metric Results

Results → accuracy degrades gracefully with motion

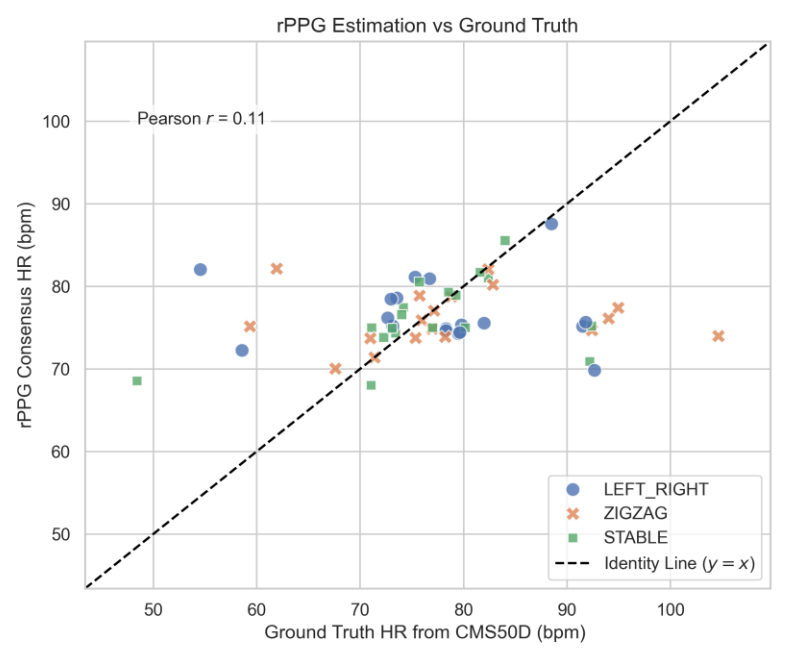
Mean MAE	Median MAE	Max MAE	Pass rate
6.93	3.29	28.03	68.5%
Outlier - Sensitive	Typical case	Worst case	

Interpretation

- Static recordings provide upper-bound performance.
- Motion increases estimation error, but reliable HR recovery is preserved.



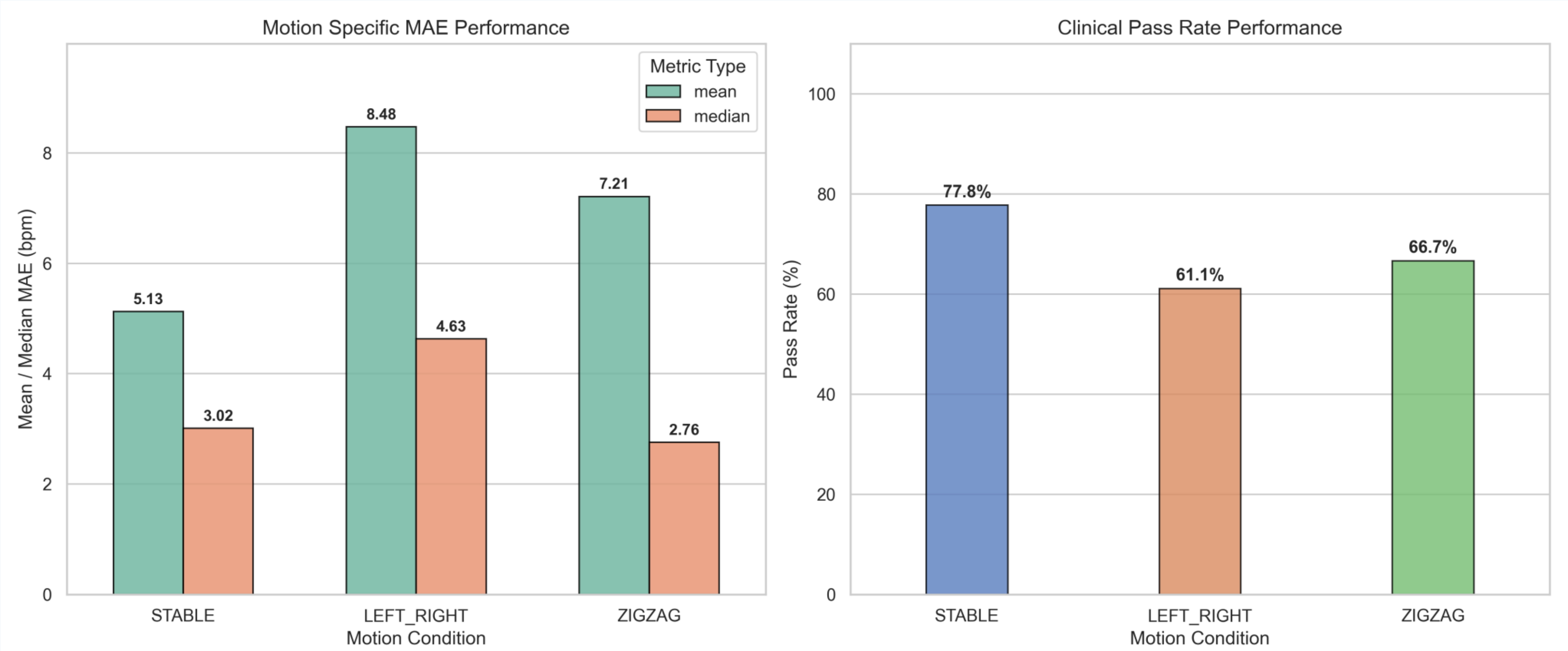
(a)



(b)

Figure 6 : (a) Bland-Altman Agreement Analysis plot (b) Estimated HR against Ground Truth Scatter Plot

Motion-Wise Performance:



(a) (b)

Figure 7 : (a) Motion Specific MAE Performance denoted using the mean and median values (b) Motion Specific Clinical Pass Rate Performance

Subject-Level Error Audit: The Main Weakness in the Dataset

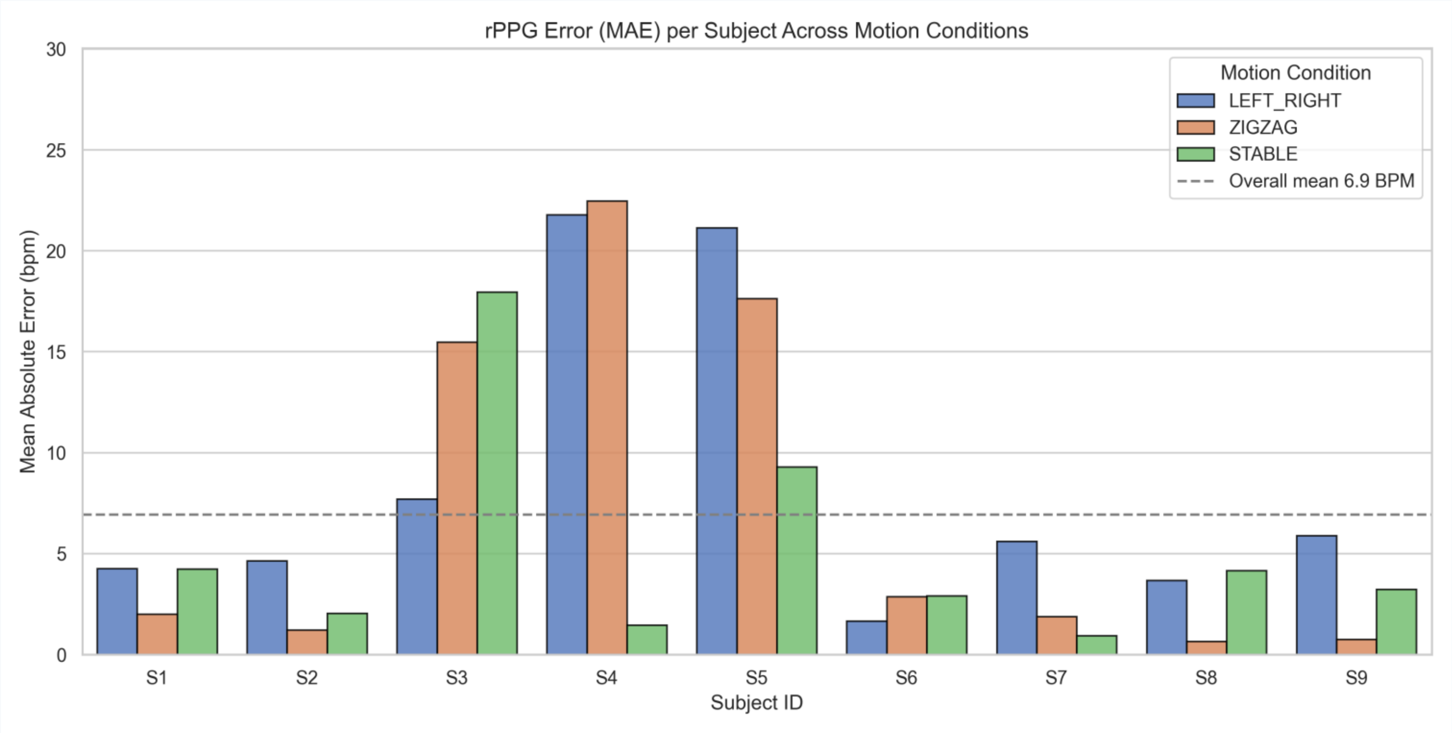


Figure 8 : rPPG error (MAE) per subject across the motion conditions with the overall mean

Best-performing subjects

S6:2.47 BPM / S2: 2.62 BPM / S7 : 2.80 BPM

Highest – error subjects

S5:16.01 BPM / S4:15.23 BPM / S3 : 13.71 BPM

Interpretation

The algorithm works well for several subjects, but generalization is a bit limited with respect to cases with outlier attributes.

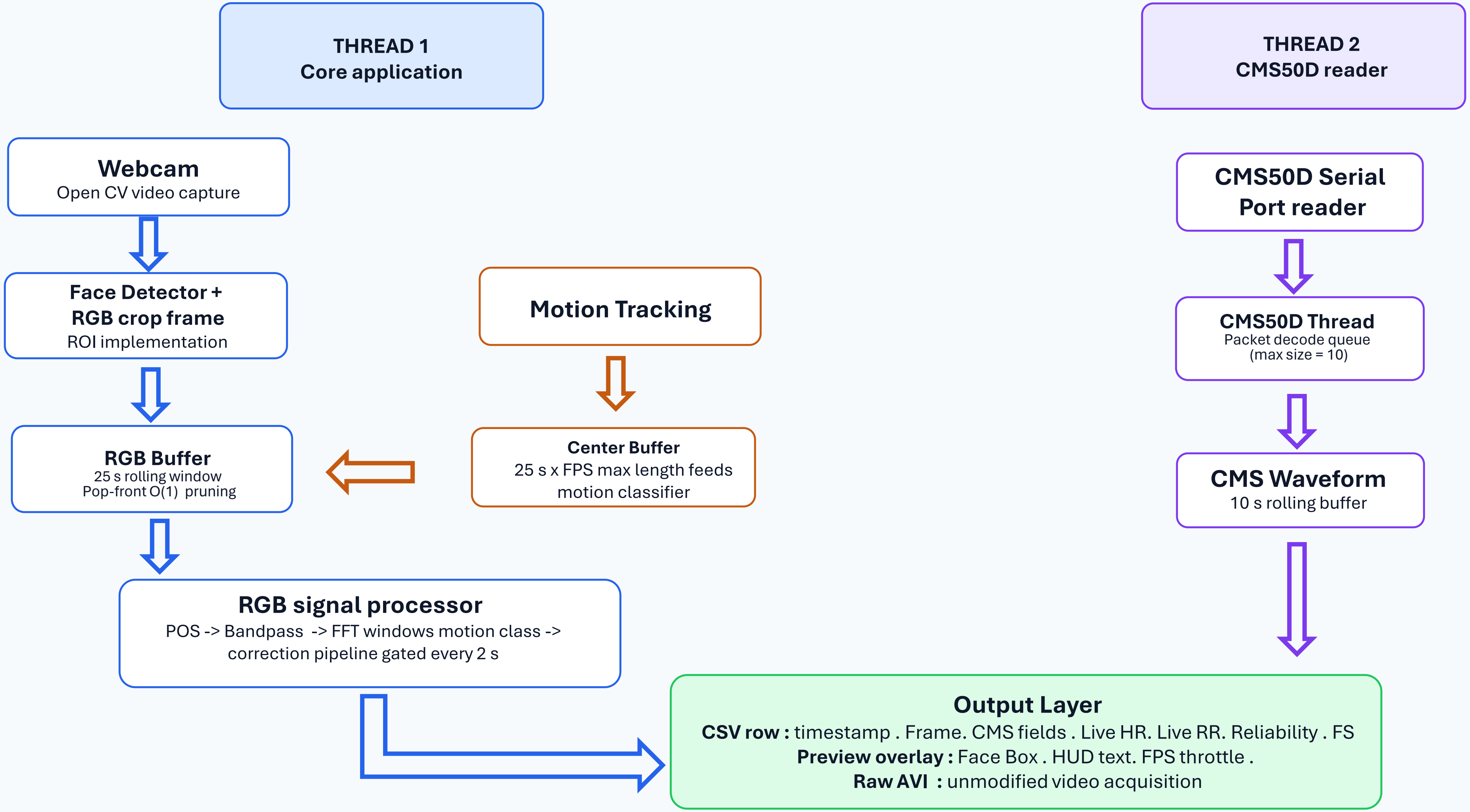
PHASE - II

Real- Time Implementation with Concurrency Model

MULTI-THREADING

Real-Time Architecture: Concurrency Model

Decouple acquisition from analysis



Handling Frame Jitter: Time-Delta Arrays

Measured timing → dynamic frequency bins

Static constant removed

- The live code avoids assuming fs = 30 Hz.
- Camera capture intervals are measured directly.
- Loop pacing absorbs slow frames, so a frame that took longer than budget doesn't compound into permanent drift.

Parallel timestamp buffers

- The camera frame timestamp and hardware timestamps are stored separately.
- Each analysis window uses its own measured duration.
- This anchors both to true time instead of to camera frame count, which allows them to stay synchronized

Frequency mapping

- HR_FFT estimates become aligned with true elapsed time.
- For RR, irregular inter-beat intervals are first interpolated onto a uniform time grid before PSD.
- A separate jitter correction step, since respiration peaks are unevenly spaced

Memory Optimization: Rolling Buffer Management

Rolling windows → stable estimates

RGB window
25 s
Live RGB window seconds

HR Gate
12 s
minimum HR window

Update rate
2 s
Live RGB window seconds

RR Gate
20 s
minimum HR window

Why it matters

- Short buffers can create false spectral peaks.
- HR gate protection prevents erratic readings being shown as real values.
- Only 3 element RGB mean vector is kept per frame, thereby keeps the memory flat regardless of resolution or duration

Real-Time Adaptive Execution Loop

Motion label → adaptive parameter switching

20-frames head-center history → Estimate motion class → Select pipeline configuration → Update filtering + ROI logic

Motion classes

- **Warmning_up :** Fallback path until enough tracking data exist.
- **Left_Right:** Yaw-dominant motion profile.
- **Zigzag :** Combined motion requiring stronger safeguards.
- **Stable:** Minimal to no head motion

Respiration Rate Safety Guardrails

Reject fragile RR estimates

Short-buffer vulnerability

- Transient facial motion can distort peak timing.
- A poor tachogram can produce unrealistic respiration rates.
- RR must be gated more strictly than HR.
- False respiration rates are more clinically risky to display confidently than a missing HR.

Biological bounding

- Respiration rate output clamped to 10–40 breaths/min.
- It can never report a number outside the physiological range.
- This prevents noise from being presented as physiology.

Hardware Integration: CMS50D Pulse Oximeter Track

Camera rPPG <~> hardware reference

CMS50D serial port → Raw byte stream parsing → Pulse waveform channel → Hardware HR

Claim as Ground Truth

- Though it is just a hardware reference, we have considered here it to be the ground truth.
- It supports cross-checking the camera rPPG output during live trials.
- Clinical-grade validation would require a stronger reference protocol.

Preview of the online HR estimation system

Raw pixels stay clean

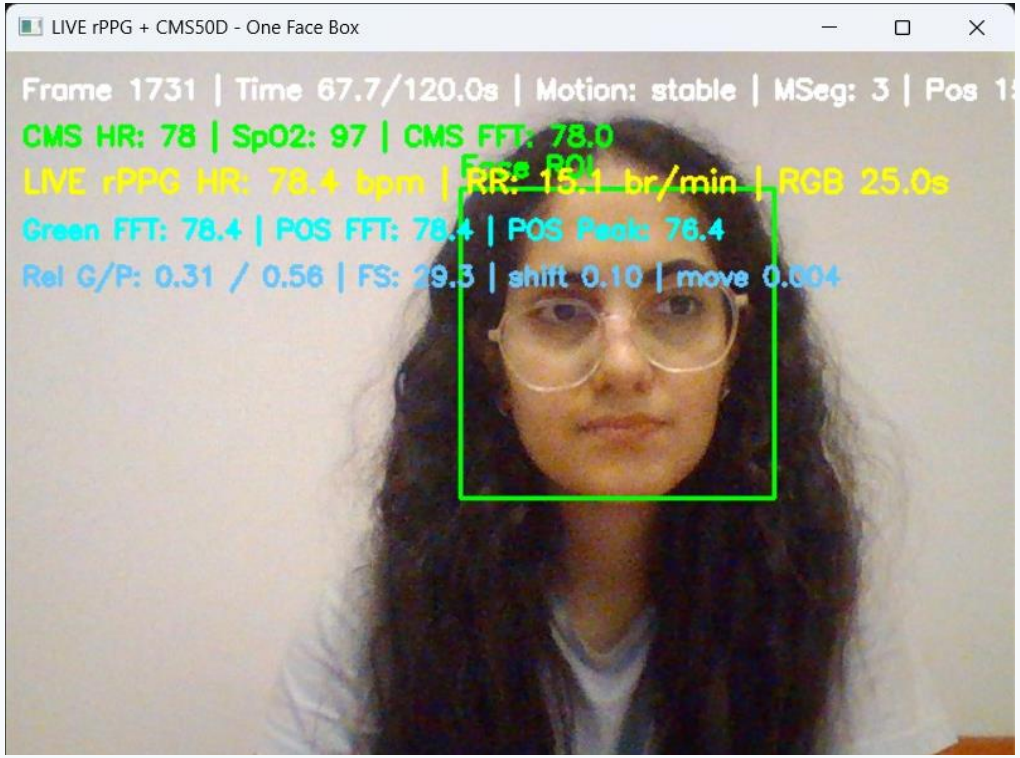


Figure 9 : Visual Feedback

Limitation with Extreme lighting

Raw pixels stay clean

Uncontrolled Lighting

MAE : 8.4 BPM

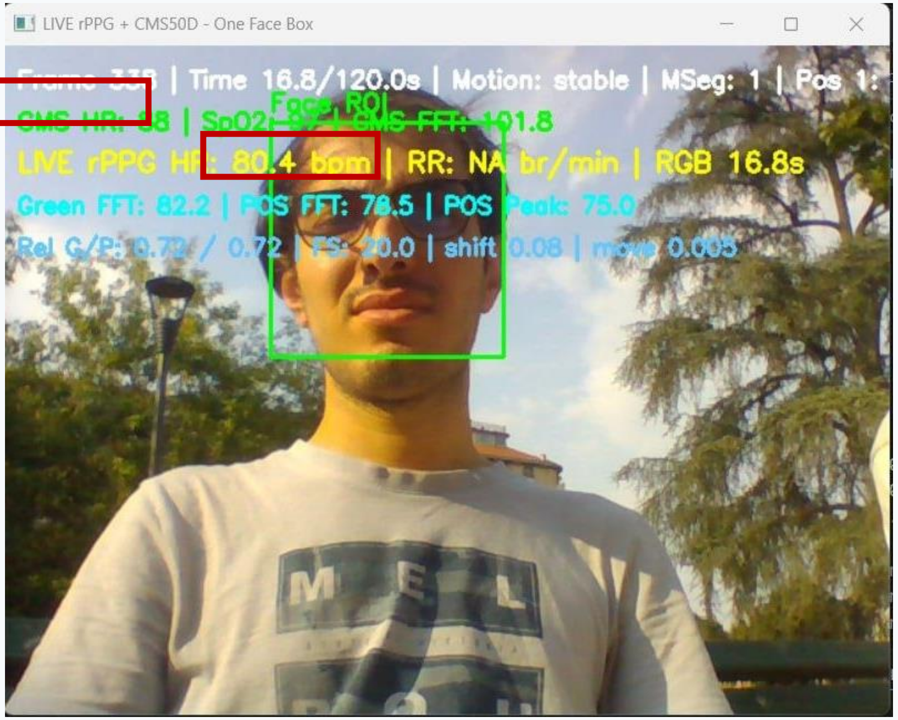


Figure 10 : Uncontrolled Lighting conditions

Controlled Lighting

MAE : 0.7 BPM

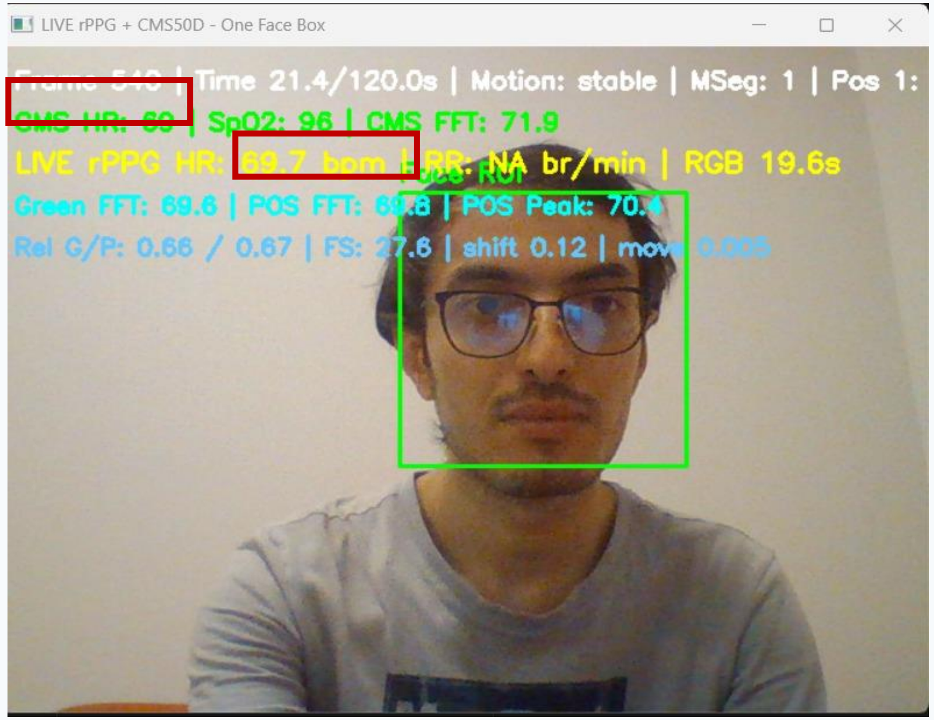


Figure 11 : Controlled Lighting conditions

Preview of the GUI

Raw pixels stay clean

Implementation

- The online processing code was then built into a web based application with the backend developed on python and Fastapi.
- While the frontend was developed using typescript and vite react.

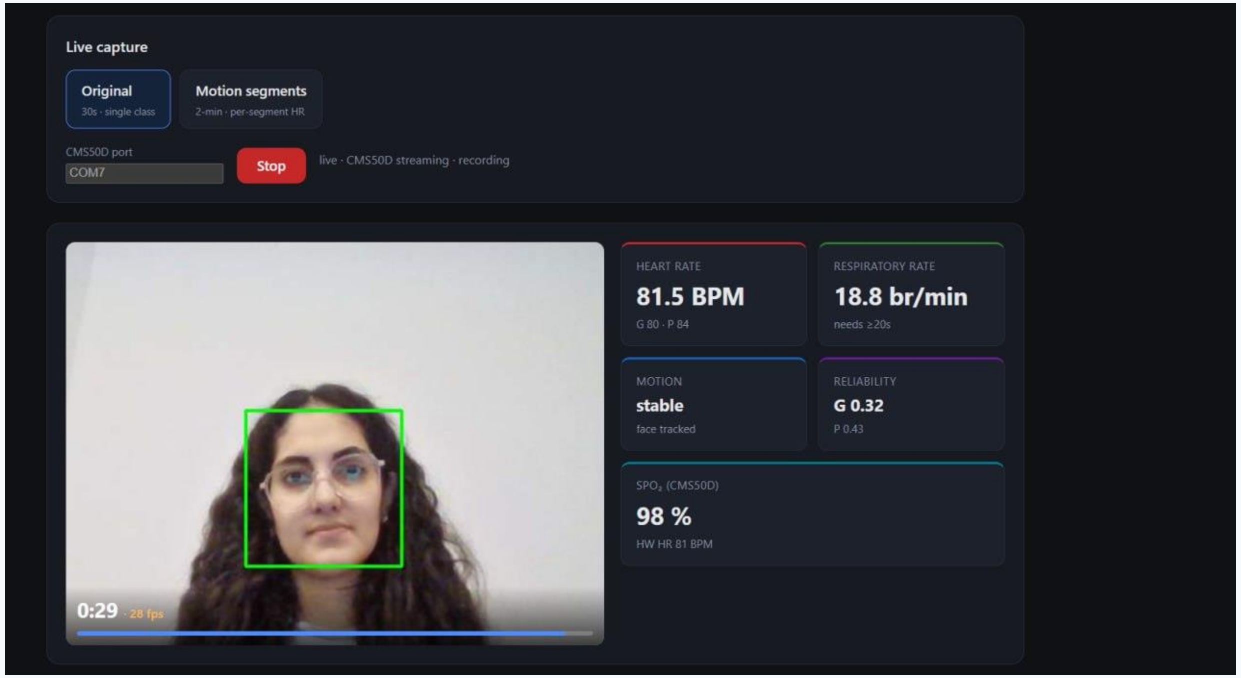


Figure 12 : GUI of the application

Thank you for your attention!